PROJECT OVERVIEW

Project: Distributed Splunk SOC Lab – Setup & Configuration

Project Scenario

This project involves designing and implementing a distributed Security Operations Center (SOC) lab using Splunk Enterprise across multiple systems.

The lab simulates a centralized security log collection and monitoring environment, where logs generated from Linux and Windows systems are forwarded, indexed, and searched through a distributed Splunk architecture.

Key Activities Performed

1. Virtual Lab Environment Setup

- **Created Kali Linux, Ubuntu 01, and Ubuntu 02 virtual machines using Oracle VirtualBox.**
- **Configured NAT Network and Host-Only Network adapters.**
- **Established network connectivity between the virtual machines and Windows host.**

2. Splunk Enterprise Deployment

- **Installed Splunk Enterprise on Ubuntu 01, Ubuntu 02, and Windows.**
- **Configured Ubuntu 01 as the Heavy Forwarder.**
- **Configured Ubuntu 02 as the Indexer.**
- **Configured Windows as the Search Head.**

3. Linux Log Collection

- **Configured Kali Linux to generate and forward logs using rsyslog.**
- **Configured Ubuntu 01 to receive syslog data over TCP port 5514.**
- **Configured Ubuntu 01 to monitor its local authentication, system, and APT logs.**

4. Windows Log Collection

- **Installed Splunk Universal Forwarder on the Windows host.**
- **Configured collection of Security, System, and Application Event Logs.**
- **Forwarded Windows events to Ubuntu 01 over TCP port 9998.**

5. Centralized Log Forwarding and Indexing

- **Configured Ubuntu 01 to forward collected events to Ubuntu 02 over TCP port 9997.**
- **Created five custom Splunk indexes to organize Linux and Windows event data.**
- **Verified event ingestion and index availability.**

6. Distributed Search Configuration

- **Connected the Windows Search Head to the Ubuntu 02 Indexer.**
- **Configured distributed search through the Splunk management interface.**
- **Verified event retrieval from all five custom indexes.**

7. End-to-End Verification

- **Tested network connectivity and Splunk receiving ports.**
- **Verified forwarding connections between the components.**
- **Executed SPL searches to confirm successful event retrieval.**
- **Captured screenshots as implementation and verification evidence.**

Final Project Outcome

Established a distributed Splunk SOC lab demonstrating centralized log collection, forwarding, indexing, and searching across Linux and Windows environments.

Architecture:

Kali Linux → Ubuntu 01 (Heavy Forwarder) → Ubuntu 02 (Indexer) → Windows (Search Head) - The next implementation phase will focus on SOC detection searches, correlation searches, alerts, dashboards, and security investigation workflows.

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Distributed Splunk SOC Lab: Setup & Configuration Guide Documentation

1. Project Specifications

1.1 Project Objective and Scope

The objective of this project is to design and implement a distributed Security Operations Center (SOC) lab environment using Splunk Enterprise and multiple virtual machines.

The lab demonstrates centralized log collection, forwarding, indexing, and searching across different systems. Kali Linux is used to generate security-related logs, while Splunk Enterprise components process and store the collected events for investigation.

The project covers virtual network configuration, system connectivity, Splunk Enterprise installation, Heavy Forwarder configuration, Indexer configuration, syslog forwarding, Search Head configuration, and end-to-end verification.

1.2 Physical Host System Specifications

Component	Specification
Operating System	Windows
Processor	AMD Ryzen 7 7435 HS
Installed RAM	24 GB
Virtualization Platform	Oracle VirtualBox 7.2.22 r175245

1.3 Virtual Machine Specifications

The lab consists of three virtual machines.

VM	OS	Allocated RAM	CPU Cores	Disk Space
Kali Linux	Kali Linux - Debian 64-bit	4 GB	4	25
Ubuntu 01	Ubuntu Linux - Ubuntu 25.04	4 GB	3	30
Ubuntu 02	Ubuntu Linux - Ubuntu 25.04	6 GB	4	30

The disk figures are the virtual disk capacities displayed in VirtualBox, not necessarily the amount of physical storage currently occupied

1.4 Software and Version Details

Software	System	Version / Status
Oracle VirtualBox	Windows Host	7.2.22 r175245
Splunk Enterprise	Windows	10.4.4
Splunk Enterprise	Ubuntu 01	Pending installation
Splunk Enterprise	Ubuntu 02	Pending installation
Kali Linux	Log-generation machine	Debian-based 64-bit
Ubuntu Linux	Ubuntu 01 and Ubuntu 02	25.04
rsyslog	Kali Linux	To be verified

Splunk License: *Splunk Enterprise Download Trial — currently configured under the Trial license group*

1.5 Final Architecture and Machine Roles

The distributed lab is designed with the following machine roles:

Machine	Assigned Role	Responsibility
Kali Linux	Log Generator	Generates test and security-related logs and forwards them using rsyslog.
Ubuntu 01	Heavy Forwarder	Receives incoming logs, processes them, and forwards events to the Indexer.
Ubuntu 02	Indexer	Receives, processes, and stores incoming events in configured indexes.
Windows Host	Search Head	Provides the search interface for querying and investigating indexed events.

Planned Log Flow:

Kali Linux → Ubuntu 01 (Heavy Forwarder) → Ubuntu 02 (Indexer) → Windows (Search Head)

Note: This represents the intended architecture. Each component and communication path will be marked as verified only after successful implementation and testing.

2. Virtual Network Configuration

2.1 Virtual Network Configuration

The virtual network environment was configured using Oracle VirtualBox to provide internet access and private communication between the Windows host and the virtual machines.

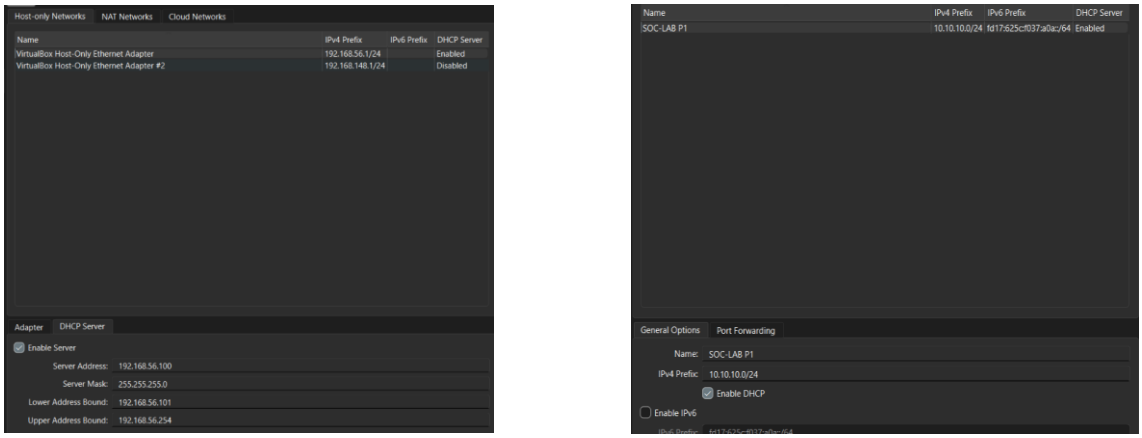
The following network configurations were created:

NAT Network

- Network Name: SOC-LAB P1
- IPv4 Network: 10.10.10.0/24
- DHCP Server: Enabled
- Purpose: To provide internet connectivity to the virtual machines for downloading packages, installing software and performing system updates.

Host-Only Network

- Network Name: VirtualBox Host-Only Ethernet Adapter
- IPv4 Network: 192.168.56.0/24
- Host IPv4 Address: 192.168.56.1
- DHCP Server: Enabled
- Purpose: To establish private network communication between the Windows host and the virtual machines.



Host-Only Adapter Configuration

2.2 Virtual Machine Network Adapter Configuration

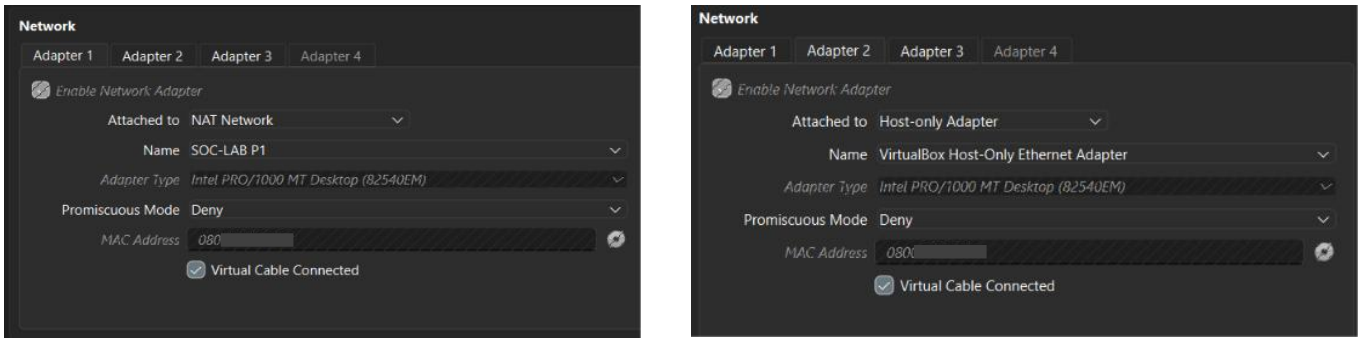
The virtual machines were configured with separate network adapters to support internet access and private lab communication.

Virtual Machine	Adapter 1	Adapter 2
Kali Linux	NAT Network — SOC-LAB P1	Host-Only Adapter
Ubuntu 01	NAT Network — SOC-LAB P1	Host-Only Adapter
Ubuntu 02	NAT Network — SOC-LAB P1	Host-Only Adapter

Purpose of the configuration:

- Adapter 1 was assigned to the NAT Network to provide internet connectivity.
- Adapter 2 was assigned to the Host-Only Network for private communication with the Windows host and other connected lab systems.

Note: The exact Host-Only adapter assignment for each VM and the Windows host interface will be confirmed during connectivity verification.



Network adapter configuration for the 3 linux distros

2.3 Linux Network Configuration and Internet Connectivity Verification

Kali Linux, Ubuntu 01 and Ubuntu 02 were configured with two network interfaces: a NAT Network adapter for internet connectivity and a Host-Only adapter for private communication with the Windows host and other virtual machines.

IP Address Allocation

Machine	NAT Interface	NAT IPv4 Address	Host-Only Interface	Host-Only IPv4 Address	Default Gateway
Kali Linux	eth0	10.10.10.3/24	eth1	192.168.56.104/24	10.10.10.1
Ubuntu 01	enp0s3	10.10.10.7/24	enp0s8	192.168.56.103/24	10.10.10.1
Ubuntu 02	enp0s3	10.10.10.6/24	enp0s8	192.168.56.105/24	10.10.10.1

Commands Used for Network Verification

The following commands were executed on all three Linux machines to identify network interfaces, IP addresses and routing information.

1. Display IP addresses and network interfaces

ip -br addr

2. Display network interface details and MAC addresses

ip link show

3. Display routing table and default gateway

ip route

Internet Connectivity Verification

External connectivity and DNS resolution were tested using the following commands on each Linux machine.

4. Test external IP connectivity

ping -c 4 8.8.8.8

5. Test DNS resolution and domain connectivity

ping -c 4 google.com

Verification Results

Machine	External IP Connectivity	DNS Resolution	Packet Loss
Kali Linux	Successful	Successful	0%
Ubuntu 01	Successful	Successful	0%
Ubuntu 02	Successful	Successful	0%

Verification Outcome: All three Linux machines successfully obtained their respective NAT Network and Host-Only IP addresses. Internet connectivity and DNS resolution were verified using ping tests, with no packet loss during the recorded tests.

3. Network Connectivity and SSH Verification

3.1 Network Interface and IP Address Verification

After configuring the NAT Network and Host-Only adapters, the network interfaces and assigned IP addresses were verified on all three Linux virtual machines.

The following commands were used to inspect the network configuration:

Display IP addresses and network interfaces

ip addr

Display network interface status and MAC addresses

ip link show

Display routing table and default gateway

ip route

The commands were executed on Kali Linux, Ubuntu 01, and Ubuntu 02 to confirm that both network interfaces were configured correctly.

3.2 Network Connectivity Verification

Connectivity between the virtual machines and the Windows host was tested using the ping command over the Host-Only network.
The following tests were performed:

- **Windows host to Kali Linux.**
- **Windows host to Ubuntu 01.**
- **Windows host to Ubuntu 02.**
- **Kali Linux to Ubuntu 01.**
- **Kali Linux to Ubuntu 02.**
- **Kali Linux to Windows host.**

All tested connections returned successful ping responses with 0% packet loss.

Connectivity Verification Results

Source	Destination	Result
Windows Host	Kali Linux	Successful
Windows Host	Ubuntu 01	Successful
Windows Host	Ubuntu 02	Successful
Kali Linux	Ubuntu 01	Successful
Kali Linux	Ubuntu 02	Successful
Kali Linux	Windows Host	Successful

3.3 Internet and DNS Connectivity Verification

Internet connectivity and DNS resolution were verified separately on the Linux virtual machines.

Test external IP connectivity

ping -c 4 8.8.8.8

Test DNS resolution and connectivity

ping -c 4 google.com

The external IP connectivity test confirmed that the Linux virtual machines could reach an external network. The DNS test confirmed that domain names could be resolved successfully.

Verification Results

Machine	External IP Connectivity	DNS Resolution	Packet Loss
Kali Linux	Successful	Successful	0%
Ubuntu 01	Successful	Successful	0%
Ubuntu 02	Successful	Successful	0%

3.4 SSH Connectivity Verification

SSH connectivity will be verified to allow remote access to the Ubuntu virtual machines from the Windows host.

The SSH service and remote login functionality will be checked before proceeding with the Splunk installation and configuration.

4. Splunk Installation and Initial Setup

4.1 Splunk Enterprise Installation Overview

Splunk Enterprise is deployed across the distributed SOC lab environment. The existing Windows installation is retained as the Search Head, while separate Splunk Enterprise instances are installed on Ubuntu 01 and Ubuntu 02.

The installation packages are obtained from the official Splunk download portal.

Official Download Link: https://www.splunk.com/en_us/download/splunk-enterprise.html

4.2 Splunk Deployment Details

Machine	Splunk Component	Installation
Windows Host	Search Head	Splunk Enterprise
Ubuntu 01	Heavy Forwarder	Splunk Enterprise
Ubuntu 02	Indexer	Splunk Enterprise
Kali Linux	Log Generator	No Splunk installation required

4.3 Backend Configuration Files

The distributed Splunk environment will primarily be configured through backend configuration files rather than performing all configuration through the Splunk Web interface.

The configuration files are maintained in:

Ubuntu 01 and Ubuntu 02:

/opt/splunk/etc/system/local/

Windows:

C:\Program Files\Splunk\etc\system\local\

The following files will be used according to the requirements of each component.

Configuration File	Purpose
server.conf	Server-level configuration
inputs.conf	Defines data inputs and receiving configurations
outputs.conf	Defines forwarding destinations
indexes.conf	Defines index creation and storage settings
props.conf	Defines event parsing and processing
transforms.conf	Defines advanced event transformations

4.4 Configuration Implementation Plan

The backend configuration files will be maintained separately according to each machine's assigned role.

Machine	Configuration Focus
Ubuntu 01 — Heavy Forwarder	Receiving Kali syslog data and forwarding events to Ubuntu 02
Ubuntu 02 — Indexer	Enabling receiving, creating indexes, and storing incoming events
Windows — Search Head	Configuring distributed search and connecting to the remote Indexer
Kali Linux — Log Generator	Configuring rsyslog forwarding to Ubuntu 01

Only the required configuration files will be created or modified for each component. Configuration changes will be validated and the Splunk service restarted when necessary.

Note: Actual configuration parameters, file contents, commands, and verification results are documented in the respective implementation sections.

4.5 Splunk Web Interface — Instance Identification Banners

To easily distinguish the three Splunk instances in the distributed environment, custom identification banners were configured in Splunk Web.

Each instance displays a green banner indicating its assigned role.

4.5.1 Configuration File Location

The following configuration file was created on each Splunk instance:

Ubuntu 01 and Ubuntu 02:

/opt/splunk/etc/system/local/global-banner.conf

Windows:

C:\Program Files\Splunk\etc\system\local\global-banner.conf

4.5.2 Banner Configuration

The following configuration was applied separately to each instance, changing the message according to its assigned role.

Ubuntu 01 — Heavy Forwarder

```
[BANNER_MESSAGE_SINGLETON]
global_banner.visible = true
global_banner.message = HEAVY FORWARDER
global_banner.background_color = green
```

Ubuntu 02 — Indexer

```
[BANNER_MESSAGE_SINGLETON]
global_banner.visible = true
global_banner.message = INDEXER
global_banner.background_color = green
```

Windows — Search Head

```
[BANNER_MESSAGE_SINGLETON]
global_banner.visible = true
global_banner.message = SEARCH HEAD
global_banner.background_color = green
```

4.5.3 Applying Configuration Changes

After saving the configuration files, Splunk was restarted to apply the changes.

Ubuntu 01 and Ubuntu 02:

sudo -u splunk /opt/splunk/bin/splunk restart

Windows PowerShell (Run as Administrator):

Restart-Service Splunkd

4.5.4 Verification Results

After restarting the Splunk instances, the Splunk Web interfaces were refreshed to verify the displayed banners.

Machine	Assigned Role	Banner Status
Ubuntu 01	Heavy Forwarder	Verified
Ubuntu 02	Indexer	Verified
Windows	Search Head	Verified

Result: Each Splunk Web interface displays its corresponding role identification banner, making the three instances easier to distinguish during configuration and monitoring.

5. Distributed Splunk Configuration and Implementation Workflow

5.1 Implementation Overview

After completing the network setup and Splunk installation, the next stage is to configure each machine according to its assigned role in the distributed SOC architecture.

The configuration will establish a complete log collection, forwarding, indexing, and search workflow.

5.2 Log Processing Workflow

The following workflow represents the intended log flow:

1. **Kali Linux — Log Generation: Generates security and system log events and forwards them through rsyslog.**
2. **Ubuntu 01 — Heavy Forwarder: Receives the syslog events and forwards them to the Indexer.**
3. **Ubuntu 02 — Indexer: Receives incoming events, organizes them into separate indexes, and stores the data.**
4. **Windows — Search Head: Connects to the remote Indexer through distributed search and allows analysts to search and investigate the collected events.**

5.3 Configuration Sequence

The implementation will follow this order:

Order	Machine	Configuration Activity
1	Ubuntu 02	Configure receiving and create indexes
2	Ubuntu 01	Configure syslog reception and event forwarding
3	Kali Linux	Configure rsyslog and generate test logs
4	Windows	Configure distributed search and connect to the Indexer
5	All systems	Perform end-to-end verification and SOC alert testing

5.4 Expected Architecture

Kali Linux → Ubuntu 01 (Heavy Forwarder) → Ubuntu 02 (Indexer) → Windows (Search Head)

Expected Outcome: Establish a working distributed Splunk environment in which logs generated on Kali Linux are forwarded, indexed, and searchable from the Windows Search Head.

5.5. Kali Linux — Syslog Configuration

5.5.1 Rsyslog Service Verification

Kali Linux is used as the log-generation system in the distributed SOC lab. The rsyslog service is used to collect system logs and forward them to Ubuntu 01 (Heavy Forwarder).

The following command was used to verify the rsyslog service:

sudo systemctl status rsyslog

Verification Result:

The rsyslog service was found to be active and enabled. No additional installation was required.

5.5.2 Syslog Forwarding Configuration

A separate configuration file was created to define the destination for forwarding Kali Linux logs.

Configuration file location:

/etc/rsyslog.d/60-splunk-forward.conf

Command used:

sudo nano /etc/rsyslog.d/60-splunk-forward.conf

Configuration content:

. @@192.168.56.103:5514

Configuration details:

Parameter	Description
.	Forward all facilities and priorities
@@	Enable TCP-based syslog forwarding
192.168.56.103	Ubuntu 01 IP address
5514	Configured syslog destination port

5.5.3 Configuration Verification

The following command was used to verify the forwarding configuration:

cat /etc/rsyslog.d/60-splunk-forward.conf

The output confirmed that the forwarding rule was saved correctly.

Expected Log Flow:

Kali Linux (192.168.56.104) → Ubuntu 01 (192.168.56.103:5514)

Note: The forwarding rule has been configured on Kali Linux. Actual log delivery will be verified after configuring the receiving input on Ubuntu 01 and restarting rsyslog.

5.5.4 Kali Linux — Log Sources, Indexes and Sourcetypes

1. Log Generation

Kali Linux acts as the log-generation system in the SOC lab. Apache HTTP Server generates web access and error logs, while the Linux operating system generates system and authentication logs.

2. Log Files

Log source	File path
Apache Access	/var/log/apache2/access.log
Apache Error	/var/log/apache2/error.log
Linux System	System syslog
Linux Authentication	Authentication logs

3. Rsyslog Configuration

Configuration file: /etc/rsyslog.d/60-splunk-forward.conf

Apache access and error logs are monitored using the rsyslog imfile module.

- apache_access: identifies Apache access events.
- apache_error: identifies Apache error events.
- Destination: 192.168.56.103
- Protocol: TCP
- Port: 5514

4. Current Splunk Input Mapping

Ubuntu 01 Heavy Forwarder uses the following configuration:

```
[tcp://5514]
disabled = 0
index = linux_system
sourcetype = syslog
connection_host = ip
```

All events received through this input currently use the linux_system index and syslog sourcetype.

5. Proposed Index and Sourcetype Structure

Log source	Index	Sourcetype
Linux Authentication	linux_auth	linux_secure
Linux System	linux_system	linux_syslog
Apache Access	web_logs	apache
Apache Error	web_logs	apache

6. Default Index

Splunk uses main as its default index when another index is not explicitly specified. The current Ubuntu 01 input explicitly assigns events to linux_system.

Note: Index creation and final sourcetype classification will be completed during the Ubuntu 02 Indexer configuration stage.

5.6 Ubuntu 01 — Heavy Forwarder Configuration

5.6.1 System Details

Ubuntu 01 is configured as the Splunk Heavy Forwarder (HF). It receives logs from Kali Linux through rsyslog and forwards collected data to Ubuntu 02, which acts as the Indexer.

Parameter	Configuration
Hostname	Ubuntus01
IP Address	192.168.56.103
Splunk Version	10.4.4
Splunk Installation Path	/opt/splunk
Receiving Port	5514
Forwarding Destination	192.168.56.105:9997
Splunk Web Port	8000
Management Port	8089

5.6.2 Splunk Installation and Service Configuration

Splunk Enterprise 10.4.4 was installed on Ubuntu 01 using the official Debian package.

The Splunk installation directory is:

/opt/splunk

A dedicated splunk user was configured to manage the Splunk installation and its files.

Splunk was started using:

sudo -u splunk /opt/splunk/bin/splunk start --accept-license

++

Boot-start was enabled:

sudo /opt/splunk/bin/splunk enable boot-start -user splunk

The Splunk Web interface was accessed at:

http://192.168.56.103:8000

5.6.3 Configure TCP Input for Kali Linux

A TCP input was configured to receive incoming syslog messages from Kali Linux.

Configuration file:

/opt/splunk/etc/system/local/inputs.conf

Configuration:

```
[tcp://5514]
disabled = 0
index = linux_system
sourcetype = syslog
connection_host = ip
```

Configuration details:

- TCP port 5514 receives incoming log messages.
- The input is enabled.
- Incoming events are assigned to the linux_system index.
- The sourcetype is set to syslog.
- The sender's IP address is used as the host field**.**

5.6.4 Configure Local Ubuntu Log Monitoring

Ubuntu 01 was also configured to monitor its own authentication and system logs.

The following configuration was added to inputs.conf:

```
[monitor:///var/log/auth.log]
disabled = 0
index = linux_auth
sourcetype = linux_secure

[monitor:///var/log/syslog]
disabled = 0
index = linux_system
sourcetype = linux_syslog
```

The monitored log files are:

Log file	Index	Sourcetype
/var/log/auth.log	linux_auth	linux_secure
/var/log/syslog	linux_system	linux_syslog

The Splunk CLI command list monitor was used to verify that both files were registered for monitoring.

5.6.5 Configure Forwarding to Ubuntu 02

Ubuntu 01 was configured to forward collected events to the Splunk Indexer running on Ubuntu 02.

Configuration file:

/opt/splunk/etc/system/local/outputs.conf

Configuration:

```
[tcpout]
defaultGroup = indexer_group

[tcpout:indexer_group]
server = 192.168.56.105:9997
```

The forwarding destination is:

- **Destination IP: 192.168.56.105**
- **Destination port: 9997**
- **Protocol: Splunk TCP forwarding**

5.6.6 Verify TCP Input and Log Reception

The following command was used to inspect the active listening ports:

sudo ss -lntp | grep -E '5514|8000|8089|9997'

Ubuntu 01 was confirmed to be listening on TCP port 5514.

The effective Splunk input configuration was inspected using:

sudo -u splunk /opt/splunk/bin/splunk btool inputs list --debug

The output confirmed the configured TCP input and its associated settings.

Kali-to-Ubuntu01 Log Transmission Test

A test HTTP request was generated on Kali Linux:

curl -I http://localhost/soc-test-404

The resulting Apache access event was observed on Ubuntu 01 using:

sudo tcpdump -A -s 0 -ni any 'tcp dst port 5514'

Observed source and destination:

- Source: 192.168.56.104 — Kali Linux
- Destination: 192.168.56.103 — Ubuntu 01
- Destination port: 5514

The received payload contained:

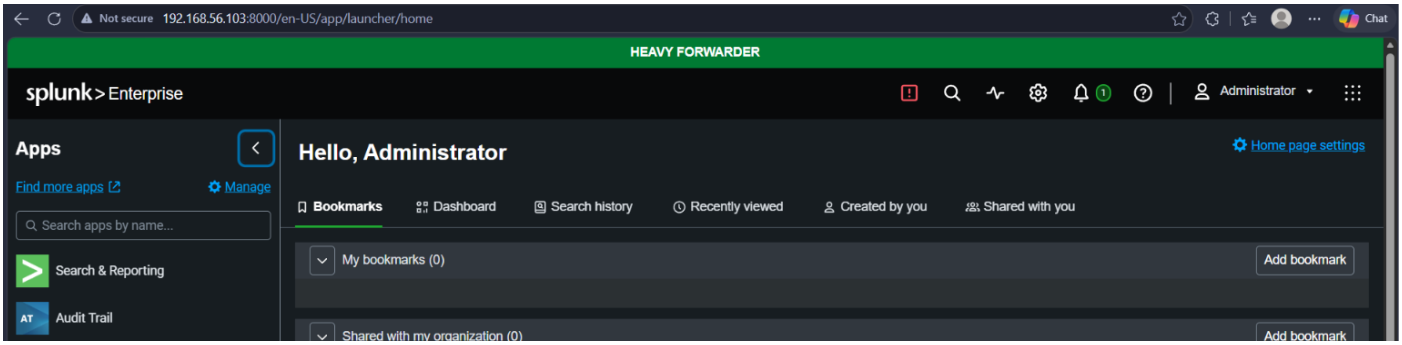
apache_access: ::1 - - [31/Oct/2026:19:32:00 +0530] "HEAD /soc-test-404 HTTP/1.1" 404 140

This confirmed that the Apache access log message generated on Kali Linux successfully reached Ubuntu 01 through the rsyslog forwarding configuration.

5.6.7 Configuration Summary

Ubuntu 01 is configured as the Heavy Forwarder with the following responsibilities:

1. Receive incoming syslog messages from Kali Linux.
2. Monitor its own authentication and system logs.
3. Assign configured indexes and sourcetypes to monitored inputs.
4. Forward collected events to Ubuntu 02 on TCP port 9997.



Ubuntus01

```
ubuntus01@Ubuntus:~$ sudo tcpdump -A -s 0 -ni any 'tcp dst port 5514'
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
14:02:01.680913 enp0s8 In  IP 192.168.56.104.53130 > 192.168.56.103.5514: Flags [P.], seq 3170119566:3
170119701, ack 4023656491, win 126, options [nop,nop,TS val 1826031509 ecr 59539814], length 135
E...^.@.@.....8h..8g....../... +...~.......
l......f<182>Oct 31 19:32:00 linus apache_access: ::1 - - [31/Oct/2026:19:32:00 +0530] "HEAD /soc-test
-404 HTTP/1.1" 404 140 "-" "curl/8.22.0"

^C
1 packet captured
5 packets received by filter
0 packets dropped by kernel
```

Ss -lntp and the tcp dump thing right

```
/opt/splunk/etc/apps/splunk_secure_gateway/default/inputs.conf        python.required = 3.13
/opt/splunk/etc/apps/splunk_secure_gateway/default/inputs.conf        run_only_one = false
/opt/splunk/etc/system/default/inputs.conf                            [tcp]
/opt/splunk/etc/system/default/inputs.conf                            _rcvbuf = 1572864
/opt/splunk/etc/system/default/inputs.conf                            acceptFrom = *
/opt/splunk/etc/system/default/inputs.conf                            connection_host = dns
/opt/splunk/etc/system/default/inputs.conf                            host = $decideOnStartup
/opt/splunk/etc/system/default/inputs.conf                            index = default
/opt/splunk/etc/system/local/inputs.conf                              [tcp://5514]
/opt/splunk/etc/system/default/inputs.conf                            _rcvbuf = 1572864
/opt/splunk/etc/system/local/inputs.conf                              connection_host = ip
/opt/splunk/etc/system/local/inputs.conf                              disabled = 0
/opt/splunk/etc/system/default/inputs.conf                            host = $decideOnStartup
/opt/splunk/etc/system/local/inputs.conf                              index = linux_system
/opt/splunk/etc/system/local/inputs.conf                              sourcetype = syslog
/opt/splunk/etc/system/default/inputs.conf                            [udp]
/opt/splunk/etc/system/default/inputs.conf                            _rcvbuf = 1572864
/opt/splunk/etc/system/default/inputs.conf                            connection_host = ip
/opt/splunk/etc/system/default/inputs.conf                            host = $decideOnStartup
/opt/splunk/etc/system/default/inputs.conf                            index = default
```

Btool

Verification status: Kali-to-Ubuntu01 TCP log delivery has been confirmed. Successful forwarding from Ubuntu 01 to Ubuntu 02 and final event indexing will be verified in the next configuration section.

5.7 Ubuntu 02 — Splunk Indexer Configuration

5.7.1 System Overview

Ubuntu02 is configured as the Splunk Enterprise Indexer in the distributed SOC lab architecture.

System details

Parameter	Configuration
Hostname	Ubuntus02
Operating System	Ubuntu 25.04
Host-only IP	192.168.56.105
NAT IP	10.10.10.6
Splunk Version	Enterprise 10.4.4
Installation Directory	/opt/splunk
Splunk Web Port	8000
Management Port	8089
Receiving Port	9997
Virtual Disk	50 GB
RAM	6144 MB

5.7.2 Splunk Enterprise Installation

Splunk Enterprise 10.4.4 was installed on Ubuntu02 using the official Linux AMD64 Debian package.

Installation directory:

/opt/splunk

A dedicated splunk system user was configured to run the Splunk service.

Directory ownership was assigned using:

sudo chown -R splunk:splunk /opt/splunk

Splunk was started using:

sudo -u splunk /opt/splunk/bin/splunk start --accept-license

Boot-start was enabled using:

sudo /opt/splunk/bin/splunk enable boot-start -user splunk

The Splunk Web interface was accessed at:

http://192.168.56.105:8000

Initial login and service operation were verified.

5.7.3 Indexer Identification

A custom global banner was configured to identify the instance as the INDEXER.

Configuration file:

/opt/splunk/etc/system/local/global-banner.conf

Configuration:

```
[BANNER_MESSAGE_SINGLETON]

global_banner.visible = true
global_banner.message = INDEXER
global_banner.background_color = green
```

The banner was displayed in Splunk Web to distinguish Ubuntu02 from the Heavy Forwarder and Search Head instances.

5.7.4 Receiving Configuration

Ubuntu02 was configured to receive forwarded Splunk data from Ubuntu01 Heavy Forwarder.

Configuration file:

/opt/splunk/etc/system/local/inputs.conf

Configuration:

[splunktcp://9997]

disabled = 0

Splunk was restarted after applying the receiving configuration.

The receiving port was verified using:

sudo ss -lntp | grep 9997

The output confirmed that Splunk was listening on TCP port 9997.

Network connectivity from Ubuntu01 was also tested:

nc -vz 192.168.56.105 9997

The connection succeeded.

5.7.5 Index Creation

Three dedicated indexes were created to organize incoming Linux log data.

Configuration file:

/opt/splunk/etc/system/local/indexes.conf

Configuration:

[linux_auth]

homePath = $SPLUNK_DB/linux_auth/db

coldPath = $SPLUNK_DB/linux_auth/colddb

thawedPath = $SPLUNK_DB/linux_auth/thaweddb

[linux_system]

homePath = $SPLUNK_DB/linux_system/db

coldPath = $SPLUNK_DB/linux_system/colddb

thawedPath = $SPLUNK_DB/linux_system/thaweddb

[linux_apt]

homePath = $SPLUNK_DB/linux_apt/db

coldPath = $SPLUNK_DB/linux_apt/colddb

thawedPath = $SPLUNK_DB/linux_apt/thaweddb

Splunk was restarted after creating the indexes.

The index configuration was verified using:

sudo -u splunk /opt/splunk/bin/splunk list index

The following indexes were confirmed:

- linux_auth
- linux_system
- linux_apt

These indexes provide separate storage locations for authentication, system, and package-management logs.

5.7.6 Ubuntu02 Local Log Forwarding

Ubuntu02's local system and authentication logs were configured to forward through Ubuntu01 Heavy Forwarder.

The forwarding mechanism uses rsyslog with the imfile module to monitor:

- /var/log/auth.log
- /var/log/syslog

Configuration file:

/etc/rsyslog.d/60-ubuntu02-forward.conf

Configuration:

```
module(load="imfile")

input(type="imfile"
    File="/var/log/auth.log"
    Tag="ubuntu02_auth:"
    Severity="info"
    Facility="local5")

input(type="imfile"
    File="/var/log/syslog"
    Tag="ubuntu02_syslog:"
    Severity="info"
    Facility="local5")

local5.* @@192.168.56.103:5514
```

The forwarding destination is Ubuntu01:

- Destination IP: 192.168.56.103
- Destination Port: 5514
- Transport Protocol: TCP

The rsyslog configuration was validated using:

sudo rsyslogd -N1

Validation completed successfully.

The rsyslog service was restarted:

sudo systemctl restart rsyslog

5.7.7 End-to-End Forwarding Verification

A test event was generated on Ubuntu02 using:

logger "UBUNTU02_FORWARD_TEST_2026"

Ubuntu01 was used to monitor incoming traffic:

sudo tcpdump -A -s 0 -ni any 'tcp dst port 5514'

The test message was observed in the captured network payload.

This verified the forwarding path:

Ubuntu02 → Ubuntu01 Heavy Forwarder → Ubuntu02 Indexer

5.7.8 Splunk Event Indexing Verification

The test event was searched in Splunk Web on Ubuntu02 using:

index=linux_system "UBUNTU02_FORWARD_TEST_2026"

The search returned 40 matching events.

The event details showed:

Field	Observed value
Host	Ubuntus02
Source	tcp:5514
Sourcetype	syslog
Index	linux_system
Test Message	UBUNTU02_FORWARD_TEST_2026

The highlighted test message confirmed that Ubuntu02's local log forwarding path was operational and that the test event was searchable in the linux_system index.

Note: The search returned 40 matching events, and the captured syslog payload showed repeated nested ubuntu02_syslog tags. This behavior should be investigated and cleaned up before treating the logging configuration as production-ready. The test confirms event delivery and indexing, not that the complete log stream is free of duplicates.

5.7.8 Virtual Disk Expansion

During search verification, Splunk displayed a minimum free disk space warning requiring 5000 MB of available storage.

Initial disk configuration:

- Virtual disk: 25 GB
- Used: 19 GB
- Available: 4.7 GB
- Usage: 80%

The VirtualBox virtual disk was increased from 25 GB to 50 GB.

The updated disk was identified using:

lsblk

The root partition was expanded using:

sudo growpart /dev/sda 2

The command confirmed that partition 2 was successfully changed.

The ext4 filesystem was expanded using:

sudo resize2fs /dev/sda2

The filesystem expansion completed successfully while mounted.

Final storage verification:

df -h

Final result:

Parameter	Final value
Root filesystem	50 GB
Used space	26 GB
Available space	22 GB
Disk usage	54%

Splunk Search & Reporting subsequently executed successfully, confirming that the minimum free-space restriction was resolved.

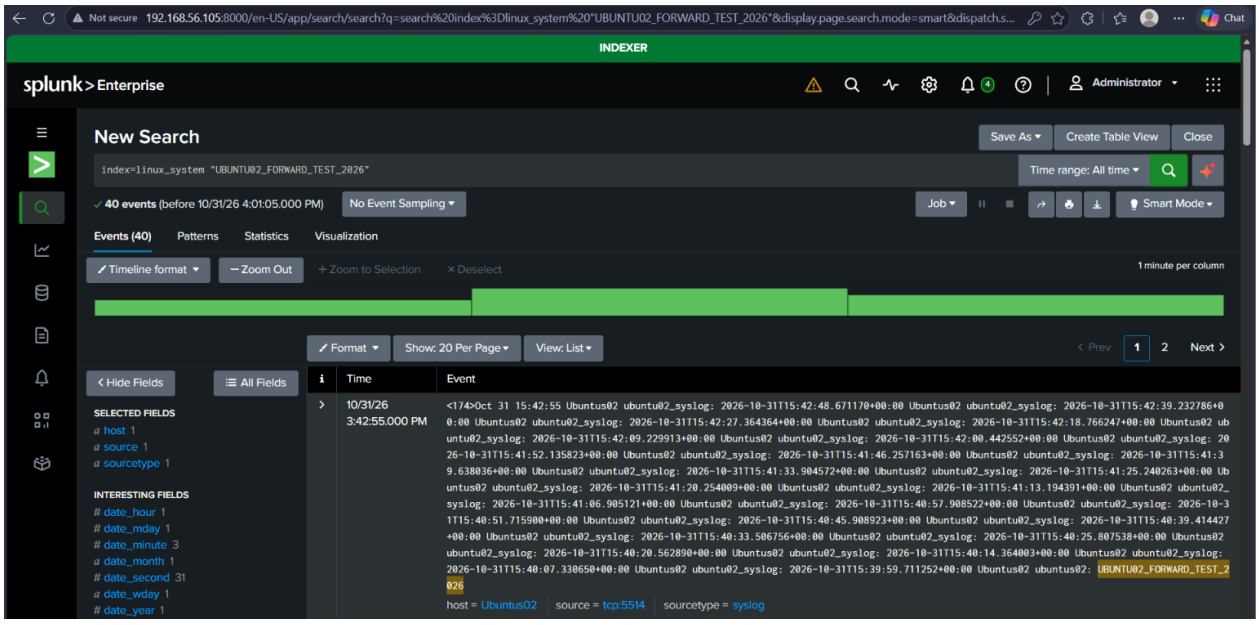
5.7.8 Final Configuration Summary

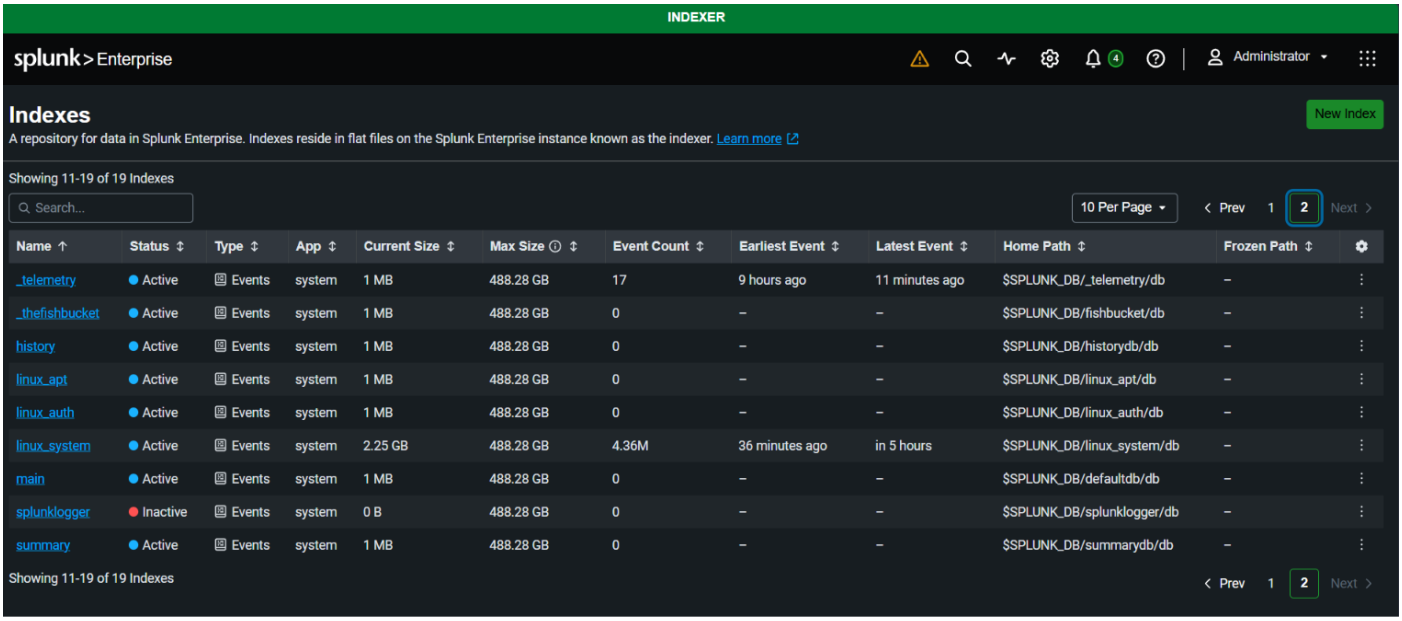
Ubuntu02 is configured with the following responsibilities:

1. Receive forwarded Splunk data from Ubuntu01 Heavy Forwarder on TCP port 9997.
2. Maintain dedicated Linux authentication, system, and package-management indexes.
3. Forward its own local authentication and system logs to Ubuntu01 through rsyslog on TCP port 5514.
4. Store incoming events in its configured Splunk indexes.
5. Provide searchable event data for the distributed Search Head.

Current verified status: Splunk Enterprise installation, receiving port, index creation, Ubuntu02-to-Ubuntu01 log forwarding, test event indexing, and storage expansion have been completed.

The next stage is configuring Windows as the Search Head and connecting it to Ubuntu02 for distributed search.





5.8. Windows — Universal Forwarder and Search Head Configuration

5.8.1 Overview

The Windows host is configured with two Splunk components:

- **Splunk Enterprise**: Acts as the Search Head for searching and analyzing indexed security events.
- **Splunk Universal Forwarder (UF):** Collects Windows Event Logs and forwards them to the Ubuntu01 Heavy Forwarder.

The Universal Forwarder sends collected events to Ubuntu01, which forwards them to Ubuntu02 for indexing.

5.8.2 Universal Forwarder Installation

The Splunk Universal Forwarder was installed on the Windows host using the Windows 64-bit MSI installer.

Installation details:

- Software: Splunk Universal Forwarder
- Version: 10.6.1.1
- Installation directory: C:\Program Files\SplunkUniversalForwarder
- Service name: SplunkForwarder

The Universal Forwarder service was verified to be running using PowerShell:

Get-Service SplunkForwarder

8.3 Forwarding Destination Configuration

During installation, the receiving indexer's destination was configured to point to the Ubuntu01 Heavy Forwarder.

Forwarding details:

Parameter	Configuration
Destination server	Ubuntu01
IP address	192.168.56.103
Receiving port	9998
Protocol	Splunk-to-Splunk

The Windows Universal Forwarder establishes a connection with the Heavy Forwarder, which subsequently relays the collected events to Ubuntu02.

5.8.4 Windows Event Log Collection

The Universal Forwarder configuration file was created at:

C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

The following inputs were configured:

```
[WinEventLog://Security]
disabled = 0
index = windows_security
renderXml = true

[WinEventLog://System]
disabled = 0
index = windows_system
renderXml = true

[WinEventLog://Application]
disabled = 0
index = windows_system
renderXml = true
```

Log collection configuration:

Windows Event Log	Splunk Index
Security	windows_security
System	windows_system
Application	windows_system

The configuration enables collection of Windows Security, System and Application events in their respective Splunk indexes.

5.8.5 Index Configuration on Ubuntu02

The following indexes were created on the Ubuntu02 Indexer to receive Windows events:

- windows_security
- windows_system

The indexes were verified using the Splunk CLI.

5.8.6 Universal Forwarder Connection Verification

The forwarding connection was verified from Windows PowerShell using:

```
& "C:\Program Files\SplunkUniversalForwarder\bin\splunk.exe" list forward-server
```

Verified forwarding status:

```
Active forwards:
   192.168.56.103:9998

Configured but inactive forwards:
   None
```

This confirmed that the Windows Universal Forwarder successfully established an active connection with the Ubuntu01 Heavy Forwarder.

5.8.7 Windows Event Generation and Ingestion Verification

A test event was generated in the Windows Application Event Log using the following PowerShell command:

```
eventcreate /T INFORMATION /ID 100 /L APPLICATION /SO SOC-LAB /D
"WINDOWS_FORWARD_TEST_2026"
```

The event was successfully created.

The following SPL search was executed to verify its arrival on the Ubuntu02 Indexer:

```
index=windows_system "WINDOWS_FORWARD_TEST_2026"
```

Verification result:

- One matching event was returned.
- Host: WIN-TPK048QP3RR
- Source: WinEventLog:Application
- Sourcetype: XmlWinEventLog:Application
- Event ID: 100
- Provider: SOC-LAB

This confirmed successful Windows Application event collection, forwarding through Ubuntu01, and indexing on Ubuntu02.

5.8.8 Windows Security and System Log Verification

Windows Security log ingestion was verified using:

```
index=windows_security
```

***The search returned approximately 312,078 events*.**

Event ID distribution was examined using:

```
index=windows_security
| stats count by EventCode
| sort - count
```

Windows System log ingestion was verified using:

```
index=windows_system
```

The search returned approximately 9,418 events.

These results confirmed that Windows Security and System events were reaching the Indexer.

5.8.9 Search Head Configuration Status

Splunk Enterprise is installed on the Windows host and is designated as the Search Head.

The distributed-search connection between the Windows Search Head and Ubuntu02 Indexer is a separate configuration task and has not yet been verified.

It will be covered in a subsequent implementation phase.

5.8.10 Section Summary

The Windows Universal Forwarder was successfully installed and configured to collect Security, System and Application Event Logs.

The forwarding connection to Ubuntu01 was verified, and the successful indexing of Windows events on Ubuntu02 was confirmed through Splunk searches.

The Windows log collection and forwarding component of the lab is operational.

6. End-to-End Configuration Verification

6.1 Overview

End-to-end verification was performed to validate communication between the lab components, Splunk forwarding connections, and successful event ingestion on the Ubuntu02 Indexer.

The verification covered Kali Linux, Ubuntu01 Heavy Forwarder, Ubuntu02 Indexer and the Windows host.

6.2 Network Connectivity Verification

The VirtualBox Host-Only network was used for communication between the lab systems.

The following IP addresses were configured:

System	IP Address
Windows Host	192.168.56.1
Ubuntu01 — Heavy Forwarder	192.168.56.103
Kali Linux	192.168.56.104
Ubuntu02 — Indexer	192.168.56.105

Network connectivity between the lab systems was tested using ICMP ping commands and TCP port connectivity checks.

The required communication paths were verified during the network configuration phase.

6.3 Splunk Receiving Port Verification

The required Splunk receiving ports were checked on the respective Ubuntu servers.

Server	Port	Purpose
Ubuntu01 — HF	5514	Syslog forwarding from Linux systems
Ubuntu01 — HF	9998	Windows Universal Forwarder connection
Ubuntu02 — Indexer	9997	Splunk-to-Splunk event receiving

The listening ports were verified using:

sudo ss -lntp

The required receiving services were confirmed to be listening.

6.4 Forwarding Connection Verification

Ubuntu01 to Ubuntu02

The forwarding connection from the Heavy Forwarder to the Indexer was verified using:

sudo -u splunk /opt/splunk/bin/splunk list forward-server

Verified output:

```
Active forwards:
   192.168.56.105:9997

Configured but inactive forwards:
   None
```

Windows Universal Forwarder to Ubuntu01

The Windows forwarding connection was verified using PowerShell:

& "C:\Program Files\SplunkUniversalForwarder\bin\splunk.exe" list forward-server

Verified output:

Active forwards:
  192.168.56.103:9998

Configured but inactive forwards:
  None

Both forwarding connections were successfully established.

6.5 Event Ingestion Verification

The following searches were executed on the Splunk Search Head to verify the presence of events in the configured indexes.

6.5.1 Linux Log Verification

Linux system log ingestion was verified using:

index=linux_system

Ubuntu authentication events were checked using:

index=linux_auth

These searches were used to verify Linux log ingestion from the configured Linux sources.

6.5.2 Windows Application Log Verification

A test event was generated on the Windows host and searched on the Indexer using:

index=windows_system "WINDOWS_FORWARD_TEST_2026"

The test event was successfully retrieved.

Verified event details:

- Host: WIN-TPK048QP3RR
- Source: WinEventLog
- Sourcetype: XmlWinEventLog
- Event ID: 100
- Provider: SOC-LAB

This confirmed successful Windows Application event collection, forwarding and indexing.

6.5.3 Windows Security Log Verification

Windows Security events were verified using:

index=windows_security

The search returned approximately 312,078 events.

Event distribution was examined using:

index=windows_security
| stats count by EventCode
| sort – count

Multiple Windows Security Event IDs were successfully retrieved.

6.5.4 Windows System Log Verification

Windows System events were verified using:

index=windows_system

The search returned approximately 9,418 events.
The returned events included Windows System Event ID 7031.
This confirmed successful ingestion of Windows System events.

6.5.5 Ubuntu02 Local Log Forwarding Verification

A test message was generated on Ubuntu02:

logger "UBUNTU02_FORWARD_TEST_2026"

The event was subsequently searched on the Indexer using:

index=linux_system "UBUNTU02_FORWARD_TEST_2026"

The test event was successfully retrieved, confirming the forwarding of the Ubuntu02 test message through Ubuntu01.

6.6 Configured Splunk Indexes

The following five custom indexes were created on the Ubuntu02 Indexer to organize and store the collected Windows and Linux event data.

Index Name	Data Source	Description
linux_auth	Ubuntu Linux systems	Stores Linux authentication logs.
linux_system	Kali Linux and Ubuntu systems	Stores Linux system and syslog events.
linux_apt	Ubuntu Linux	Stores APT package-management logs.
windows_security	Windows Host	Stores Windows Security Event Logs.
windows_system	Windows Host	Stores Windows System and Application Event Logs.

The indexes were created and verified on the Ubuntu02 Indexer using the Splunk CLI.

The following SPL searches were used to verify event availability in each index:

index=linux_auth
index=linux_system
index=linux_apt
index=windows_security
index=windows_system

Each index provides a separate logical storage location for its corresponding event data, supporting organized searching, monitoring, and security investigations.

6.7 Final Verification Summary

The following table summarizes the completed verification activities across the distributed Splunk SOC lab.

Verification Activity	Result
Virtual network connectivity	**Verified**
Ubuntu01 receiving port 5514	**Verified**
Ubuntu01 receiving port 9998	**Verified**
Ubuntu02 receiving port 9997	**Verified**
Ubuntu01 to Ubuntu02 forwarding	**Active**

Windows UF to Ubuntu01 forwarding	**Active**
Windows Application event ingestion	**Verified**
Windows Security event ingestion	**Verified**
Windows System event ingestion	**Verified**
Ubuntu02 local test event ingestion	**Verified**
Five custom indexes created	**Verified**
Windows Search Head to Ubuntu02 distributed-search connection	**Configured and verified**
Kali Linux log forwarding	**Previously verified**

The verification results demonstrate that the main network communication paths, log forwarding connections, index configuration, and event ingestion processes were successfully implemented.

6.8 Project Implementation Status

The core Splunk installation, receiving configuration, log collection, forwarding, indexing, and distributed-search connection activities have been completed.

The lab successfully demonstrated centralized collection and indexing of events from the configured Linux and Windows sources. The Windows Search Head was also successfully connected to the Ubuntu02 Indexer through distributed search.

The following activities remain for the next implementation phase:

1. Develop SOC detection searches for identifying suspicious activities.
2. Create and test correlation searches.
3. Configure alerts for relevant security events.
4. Develop monitoring dashboards for security investigation and event visibility.
5. Perform further detection and investigation testing using the collected log data**.**

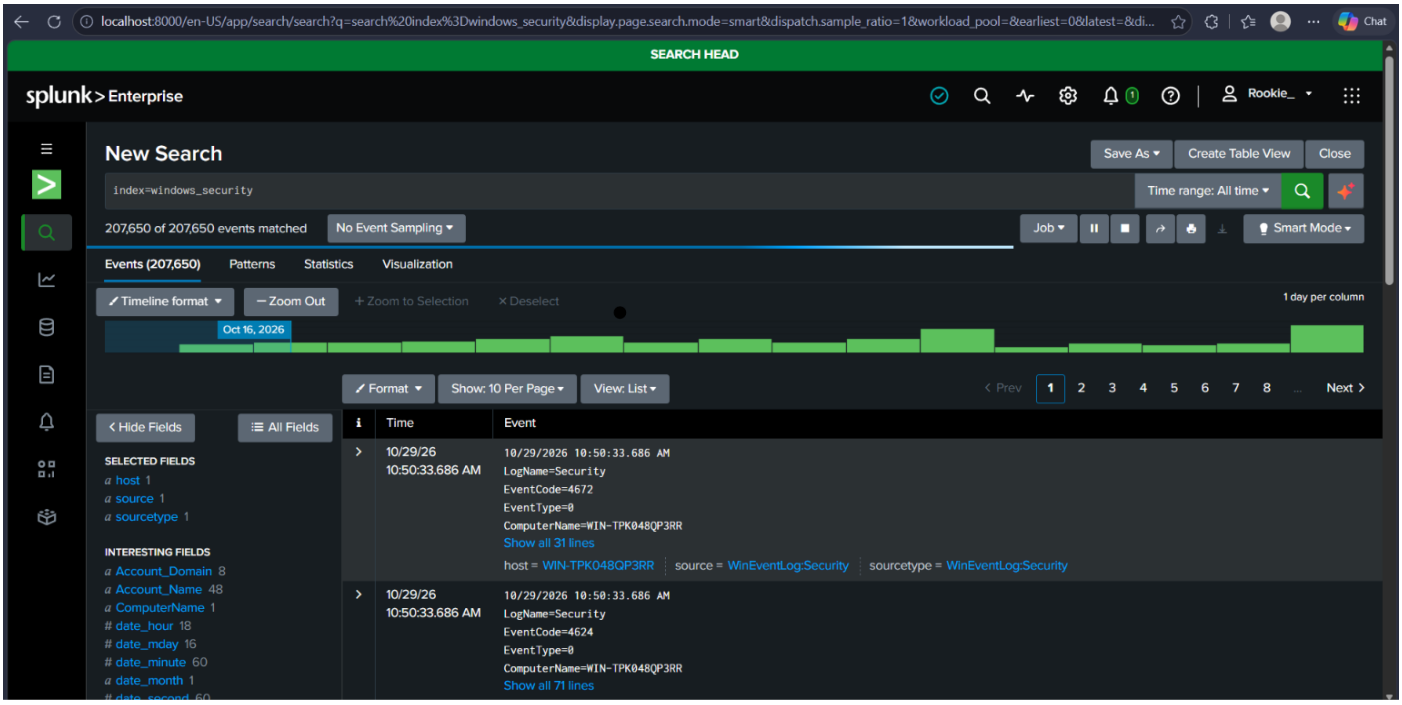
6.9 Conclusion

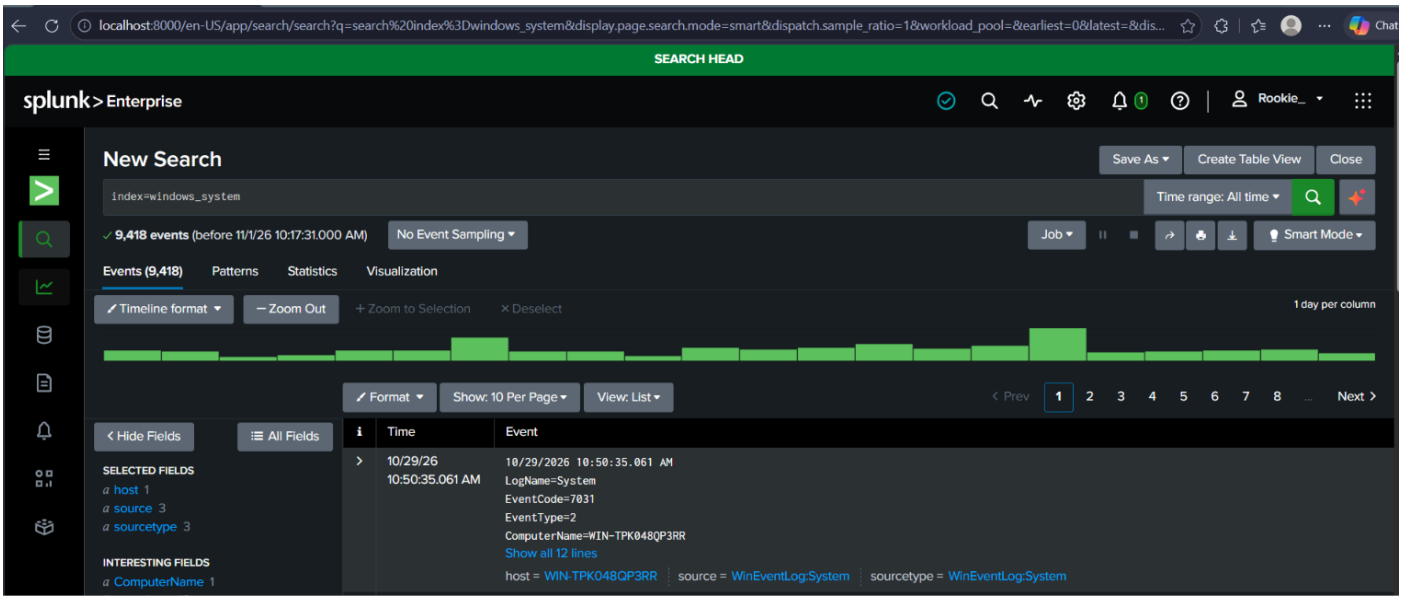
The distributed Splunk lab successfully established the core infrastructure for centralized security log collection, forwarding, indexing, and searching.

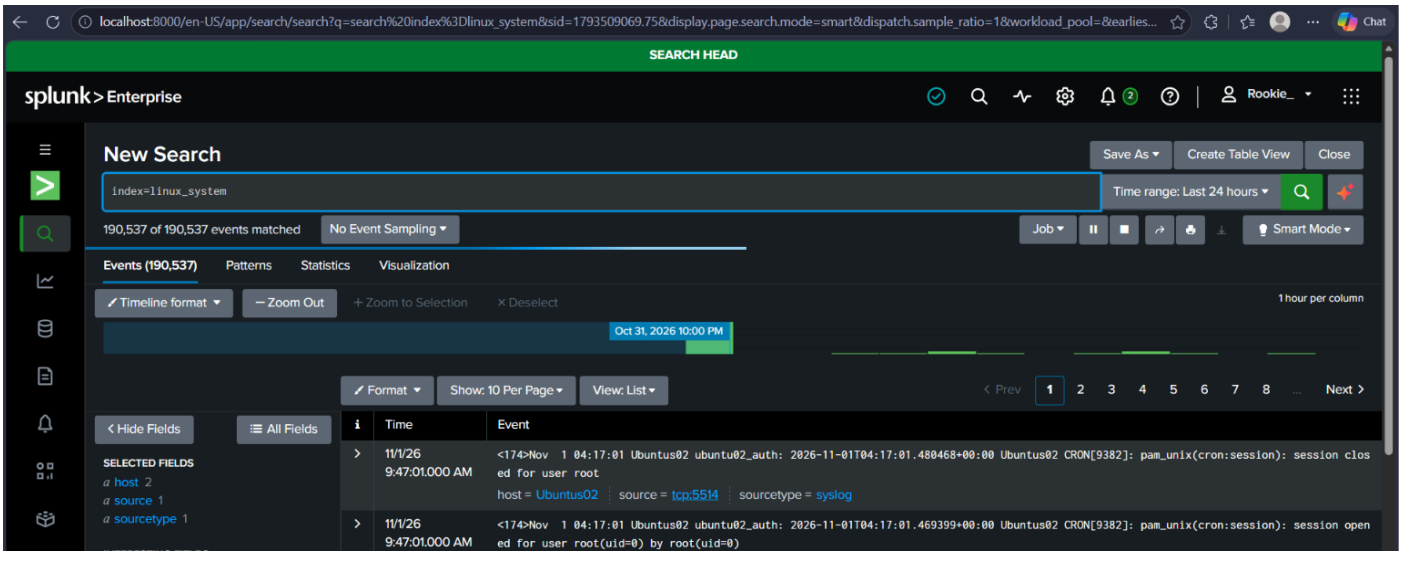
The Windows Universal Forwarder and Linux syslog forwarding mechanisms were configured to deliver events through the Ubuntu01 Heavy Forwarder to the Ubuntu02 Indexer. Five custom indexes were created to organize the collected Linux and Windows event data.

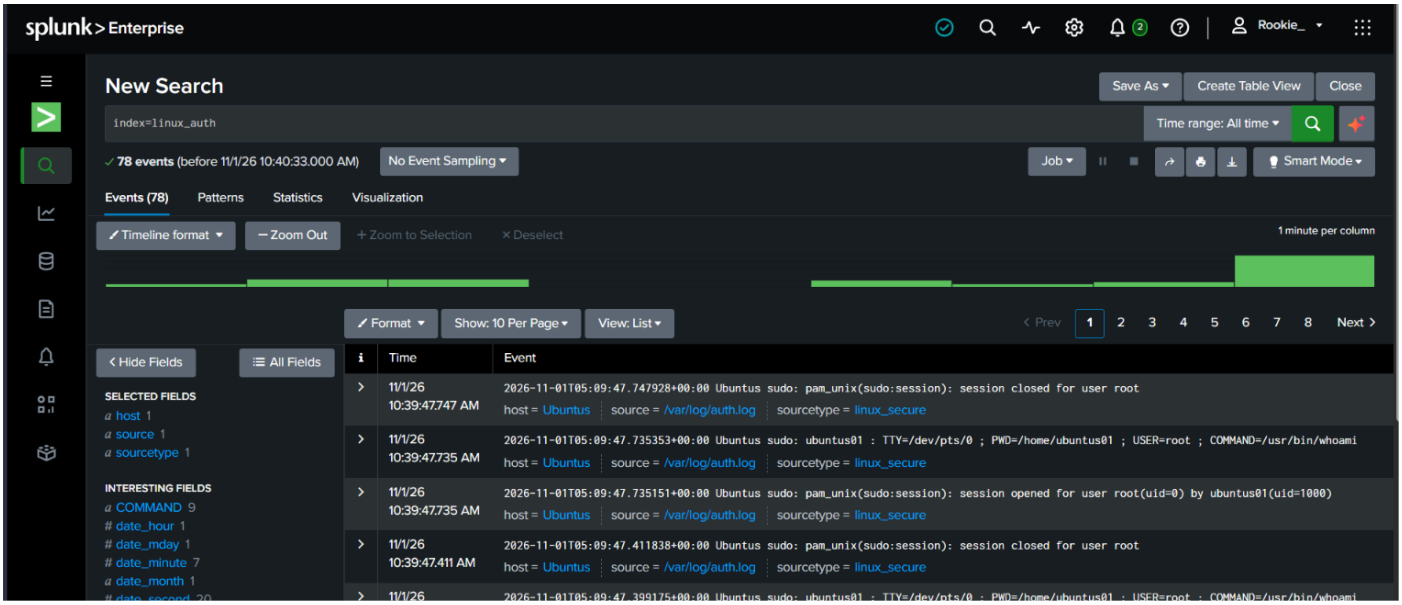
Successful event retrieval from the configured indexes demonstrated that the core collection and forwarding pipeline was operational. The Windows Search Head was also connected to the Ubuntu02 Indexer through distributed search.

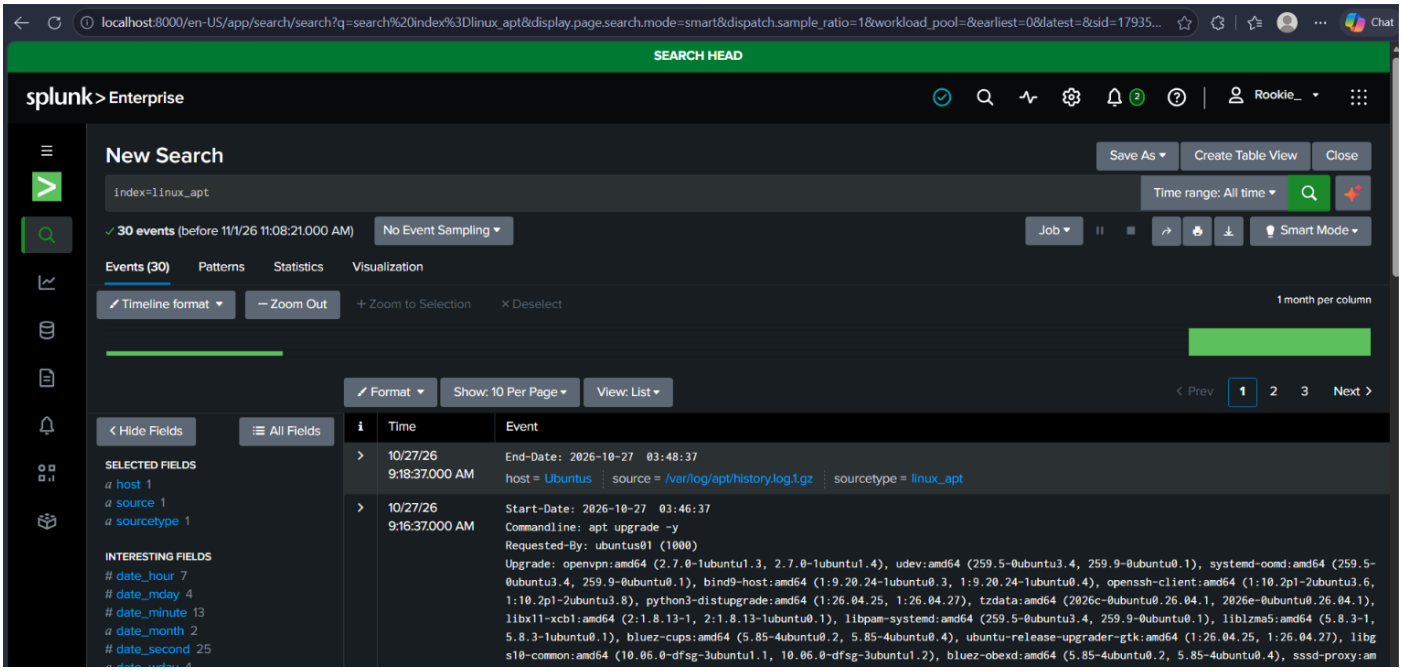
This implementation provides a foundation for the next phase of the project, focusing on centralized security monitoring, detection engineering, correlation searches, alerting, dashboards, and SOC investigation workflows.


localhost:8000/en-US/app/search/search?q=search%20index%3Dwindows_security&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earliest=0&latest=&di...
SEARCH HEAD
splunk>Enterprise
Rookie_
New Search
Save As
Create Table View
Close
index=windows_security
Time range: All time
207,650 of 207,650 events matched
No Event Sampling
Job
Smart Mode
Events (207,650)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 day per column
Oct 16, 2026
Format
Show: 10 Per Page
View: List
Prev
1 2 3 4 5 6 7 8
Next
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a Account_Domain 8
a Account_Name 48
a ComputerName 1
date_hour 18
date_mday 16
date_minute 60
a date_month 1
Time
Event
10/29/26
10:50:33.686 AM
10/29/2026 10:50:33.686 AM
LogName=Security
EventCode=4672
EventType=0
ComputerName=WIN-TPK048QP3RR
Show all 31 lines
host = WIN-TPK048QP3RR
source = WinEventLog:Security
sourcetype = WinEventLog:Security
10/29/26
10:50:33.686 AM
10/29/2026 10:50:33.686 AM
LogName=Security
EventCode=4624
EventType=0
ComputerName=WIN-TPK048QP3RR
Show all 71 lines


localhost:8000/en-US/app/search/search?q=search%20index%3Dwindows_system&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earliest=0&latest=&dis...
SEARCH HEAD
splunk>Enterprise
Rookie_
New Search
Save As
Create Table View
Close
index=windows_system
Time range: All time
9,418 events (before 11/1/26 10:17:31.000 AM)
No Event Sampling
Job
Smart Mode
Events (9,418)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 day per column
Format
Show: 10 Per Page
View: List
Prev
1 2 3 4 5 6 7 8
Next
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 3
a sourcetype 3
INTERESTING FIELDS
a ComputerName 1
Time
Event
10/29/26
10:50:35.061 AM
10/29/2026 10:50:35.061 AM
LogName=System
EventCode=7031
EventType=2
ComputerName=WIN-TPK048QP3RR
Show all 12 lines
host = WIN-TPK048QP3RR
source = WinEventLog:System
sourcetype = WinEventLog:System


localhost:8000/en-US/app/search/search?q=search%20index%3Dlinux_system&sid=1793509069.75&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earlies...
SEARCH HEAD
splunk>Enterprise
Rookie_
New Search
Save As
Create Table View
Close
index=linux_system
Time range: Last 24 hours
190,537 of 190,537 events matched
No Event Sampling
Job
Smart Mode
Events (190,537)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 hour per column
Oct 31, 2026 10:00 PM
Format
Show: 10 Per Page
View: List
Prev
1 2 3 4 5 6 7 8
Next
Hide Fields
All Fields
SELECTED FIELDS
a host 2
a source 1
a sourcetype 1
Time
Event
11/1/26
9:47:01.000 AM
<174>Nov 1 04:17:01 Ubuntus02 ubuntu02_auth: 2026-11-01T04:17:01.480468+00:00 Ubuntus02 CRON[9382]: pam_unix(cron:session): session closed for user root
host = Ubuntus02
source = tcp:5514
sourcetype = syslog
11/1/26
9:47:01.000 AM
<174>Nov 1 04:17:01 Ubuntus02 ubuntu02_auth: 2026-11-01T04:17:01.469399+00:00 Ubuntus02 CRON[9382]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)





Search Head Event Retrieval Verification

The above screenshots demonstrate successful event retrieval from all five custom indexes through the Windows Splunk Search Head. The searches confirmed the availability of Linux authentication logs (linux_auth), Linux system logs (linux_system), APT package-management logs (linux_apt), Windows Security Event Logs (windows_security), and Windows System and Application Event Logs (windows_system).

The successful retrieval of events from each index confirms that the distributed-search connection between the Windows Search Head and Ubuntu02 Indexer is operational. It also demonstrates that the collected Linux and Windows log data can be accessed and searched centrally through the Search Head.

This completes the verification of the five configured indexes and their event availability in the distributed Splunk SOC lab.